CHAPTER 5

CONTINUITY AND DIFFERENTIABILITY

CHAPTER ESSENCE AT A GLANCE:
Limits, continuity conditions, derivative rules (chain, product, quotient), implicit differentiation, logarithmic differentiation, and second-order derivatives.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Continuity at a Point:** A real function is continuous at a point if and only if the left-hand limit, the right-hand limit, and the value of the function at that point all exist and equal one another identically without breaks or jumps.

• **Differentiability and Smoothness:** Differentiability requires the existence of a unique tangent line and guarantees continuity; however, continuity does not imply differentiability at sharp corners, cusps, or vertical tangents.

• **Chain Rule and Composite Functions:** The derivative of a composite function is determined by differentiating the outer function evaluated at the inner function, multiplied by the derivative of the inner function.

• **Logarithmic and Parametric Differentiation:** Taking natural logarithms simplifies functions with variable exponents or multi-factor rational products before differentiation, while parametric curves are differentiated by dividing the parameter derivative of y by the parameter derivative of x.

ESSENTIAL FORMULAS & GOVERNING EQUATIONS

$$\lim_{x \to c^-} f(x) = \lim_{x \to c^+} f(x) = f(c)$$

Cauchy continuity criterion at point c

$$\frac{d}{dx}\left[\frac{u}{v}\right] = \frac{v\frac{du}{dx} - u\frac{dv}{dx}}{v^2}$$

Quotient rule for differentiation of algebraic fractions

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}, \qquad \frac{d^2y}{dx^2} = \frac{d}{dt}\left(\frac{dy}{dx}\right)\frac{1}{\frac{dx}{dt}}$$

First and second parametric derivatives with parameter t

$$\frac{d}{dx}\left[\sin^{-1} x\right] = \frac{1}{\sqrt{1-x^2}}, \qquad \frac{d}{dx}\left[\tan^{-1} x\right] = \frac{1}{1+x^2}$$

Standard derivatives of inverse trigonometric functions

$$y = [f(x)]^{g(x)} \rightarrow \frac{dy}{dx} = y\left[\frac{g(x)}{f(x)} f'(x) + g'(x)\ln(f(x))\right]$$

Logarithmic differentiation for variable exponential forms

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** [Trap Alert: Parametric Second Derivative] d^2y by dx^2 is NOT (d^2y by dt^2) over (d^2x by dt^2)! You MUST differentiate dy by dx with respect to t and MULTIPLY by dt by dx.

• **Alert:** [Trap Alert: Split Logarithms] When differentiating $y = u(x)^{v}(x) + p(x)^{q}(x)$, NEVER take log of both sides directly! Log of sum $\ln(u + v) \neq \ln u + \ln v$. Split into $y = u + v$.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Define continuity of a function at a point x = c. State conditions. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Definition of Continuity at a Point]** A function f(x) is said to be continuous at a point x = c in its domain if: lim {x → c} f(x) = f(c). [1 Mark]

• **[Three Necessary Conditions]** 1. f(c) is defined (exists as a finite real number). 2. lim {x → c} f(x) exists, which means Left Hand Limit equals Right Hand Limit: lim {x → c^-} f(x) = lim {x → c^+} f(x) = L. 3. The limiting value equals the functional value: L = f(c). [1 Mark]

• **[Boxed Result: LHL = RHL = f(c) for continuity at x = c]**

2. Discuss the continuity of the function f(x) = |x| at x = 0. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Setup]** f(x) = |x| = { -x for x < 0; 0 for x = 0; x for x > 0 }. Test at x = 0. [0.5 Mark]

• **[Limits Evaluation]** LHL = lim {x → 0^-} f(x) = lim {h → 0} |0 - h| = lim {h → 0} h = 0. RHL = lim {x → 0^+} f(x) = lim {h → 0} |0 + h| = lim {h → 0} h = 0. Functional value: f(0) = |0| = 0. [1 Mark]

• **[Conclusion]** Since LHL = RHL = f(0) = 0, f(x) = |x| is CONTINUOUS at x = 0. (Note: Although continuous, it is not differentiable at x = 0 due to corner point). [0.5 Mark]

• **[Boxed Result: f(x) = |x| is continuous at x = 0]**

3. Differentiate with respect to x: (a) sin(x^2 + 5), (b) cos(sin x), (c) $e^{\sin x}$. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Derivatives using Chain Rule]** (a) y = sin(x^2 + 5): dy by dx = cos(x^2 + 5) · d by dx(x^2 + 5) = 2x cos(x^2 + 5). [1 Mark]

• (b) y = cos(sin x): dy by dx = - sin(sin x) · d by dx(sin x) = - cos x · sin(sin x). [1 Mark]

• (c) y = e^(sin x): dy by dx = e^(sin x) · d by dx(sin x) = cos x · e^(sin x). [1 Mark]

• **[Boxed Result: (a) 2x cos(x^2+5) | (b) -cos x sin(sin x) | (c) cos x e^(sin x)]**

4. Differentiate with respect to x: (a) $\tan^{-1}\left(\frac{(\sin x)}{(1+\cos x)}\right)$, (b) $\sin^{-1}\left(\frac{(2x)}{(1+x\text{ffl})}\right)$. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Simplification & Differentiation]** (a) y = $\tan^{-1}$ [sin x over (1 + cos x)]: Using sin x = 2 sin(x over 2) cos(x over 2) and 1 + cos x = 2 $\cos^2$(x over 2): sin x over (1 + cos x) = tan(x over 2) ⇒ y = $\tan^{-1}$ [tan(x over 2)] = x over 2. dy by dx = d by dx(x over 2) = 1 over 2. [1.5 Marks]

• (b) y = $\sin^{-1}$ [2x over (1 + x^2)]: Substitute x = tan θ implies y = $\sin^{-1}$ [2 tan θ over (1 + $\tan^2$ θ)] = $\sin^{-1}$(sin 2θ) = 2θ = 2 $\tan^{-1}$ x. dy by dx = 2 · d by dx($\tan^{-1}$ x) = 2 over (1 + x^2). [1.5 Marks]

• **[Boxed Result: (a) dy by dx = 1 over 2 | (b) dy by dx = 2 over (1 + x^2)]**

5. Find dy by dx if x^2 + xy + y^2 = 100. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Implicit Differentiation]** Given x^2 + x y + y^2 = 100. Differentiating both sides with respect to x: d by dx(x^2) + d by dx(x y) + d by dx(y^2) = d by dx(100). [1 Mark]

• 2x + [x (dy by dx) + y(1)] + 2y (dy by dx) = 0. (x + 2y) dy by dx + (2x + y) = 0. (x + 2y) dy by dx = - (2x + y) implies dy by dx = - (2x + y) over (x + 2y). [2 Marks]

• **[Boxed Result: dy by dx = - (2x + y) over (x + 2y)]**

6. Differentiate $y = (\sin x)^x + x^{\sin x}$ with respect to x. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **[Setup]** Let y = u + v, where u = $(\sin x)^x$ and v = x^(sin x). Then dy by dx = du by dx + dv by dx. [1 Mark]

• **[Part 1: du by dx]** ln u = x ln(sin x) ⇒ (1 over u)(du by dx) = 1 · ln(sin x) + x · [cot x] = ln(sin x) + x cot x. du by dx = $(\sin x)^x$ [x cot x + ln(sin x)]. [1.5 Marks]

• **[Part 2: dv by dx]** ln v = sin x ln x ⇒ (1 over v)(dv by dx) = cos x ln x + (sin x) over x. dv by dx = x^(sin x) [(sin x) over x + cos x ln x]. [1.5 Marks]

• **[Boxed Result: dy by dx = $(\sin x)^x$ [x cot x + ln(sin x)] + x^(sin x) [(sin x) over x + cos x ln x]]**

7. If x = a(θ + sinθ) and y = a(1 - cosθ), find dy by dx at θ = π over 2. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **[Parametric Differentiation]** x = a(θ + sin θ) and y = a(1 - cos θ). dx over dθ = a(1 + cos θ) = a [2 $\cos^2$(θ over 2)]. dy over dθ = a(0 - (-sin θ)) = a sin θ = a [2 sin(θ over 2) cos(θ over 2)]. [2 Marks]

• **[Evaluating dy by dx]** dy by dx = (dy over dθ) over (dx over dθ) = [2 a sin(θ over 2) cos(θ over 2)] over [2 a $\cos^2$(θ over 2)] = tan(θ over 2). [1 Mark]

• At θ = π over 2: dy by dx = tan(π over 4) = 1. [1 Mark]

• **[Boxed Result: dy by dx = tan(θ over 2); At θ = π over 2, dy by dx = 1]**